

Your Cancer Summary Report

Michelle Thompson

Stage IIA ER+/PR+/HER2- Invasive Ductal Carcinoma with BRCA2
Germline Mutation

Precision Oncology Center — Demonstration Report

Report Date: March 26, 2026

⚠️ DRAFT - FOR CLINICIAN REVIEW

This report has not yet been reviewed by your healthcare team. Please wait for your doctor to discuss these results with you.

DRY_RUN SYNTHETIC DATA — NOT FOR CLINICAL USE. This AI-generated summary must be reviewed and approved by a licensed oncologist before any treatment decisions. Results represent synthetic demonstration data only.

At a Glance

Your Diagnosis

Breast Cancer - ER+/PR+/HER2- Invasive Ductal Carcinoma (IDC)

Stage: Stage IIA | **Grade:** Grade 2

You have been diagnosed with Stage IIA breast cancer. This is an early-to-intermediate stage breast cancer that is hormone receptor positive, meaning the cancer cells use estrogen and progesterone to grow. The good news is that this type of breast cancer responds well to hormone-blocking treatments, and your genomic testing has identified several important targets for therapy.

✓ Positive Factors

- ER+/PR+ status means the cancer responds to hormone-blocking therapies like tamoxifen
- BRCA2 germline mutation makes you eligible for PARP inhibitor therapies (olaparib, niraparib)
- PIK3CA H1047R mutation is targetable with alpelisib plus fulvestrant (FDA-approved)
- Stage IIA is early-to-intermediate stage with good treatment options
- ESR1 is highly clustered in tumor core (Moran's I 0.42) — consistent hormone-driven growth pattern
- Multi-omics analysis confirms ESR1/PGR/PIK3CA as highly significant (FDR $q < 0.001$)

⚠ Challenges to Consider

- BRCA2 germline mutation increases lifetime risk of ovarian and other cancers — family genetic counseling recommended
- MYC amplification (log2 1.45) and CCND1 amplification (log2 1.62) may contribute to treatment resistance
- CDKN2A deletion indicates loss of a key cell-cycle checkpoint
- HRD score of 35 is below the clinical threshold (42) — PARP eligibility based on BRCA2 mutation status, not HRD score alone

What Your Tests Found

Genetic testing of your tumor revealed the following important findings:

BRCA2

Pathogenic

c.5946delT frameshift variant (germline, AF=0.50)

You carry a germline (inherited) change in the BRCA2 gene that disrupts its normal function as a DNA repair gene. This mutation is responsible for your increased cancer risk but also makes your tumor more sensitive to PARP inhibitor drugs that block a backup DNA repair pathway.

Treatment Option: PARP inhibitors (olaparib, niraparib) FDA-approved for BRCA1/2-mutated breast cancer

PIK3CA

Pathogenic

H1047R missense mutation (somatic, AF=0.42, COSM775)

A change in the PIK3CA gene is activating a growth pathway in your cancer cells. This is one of the most common and well-studied mutations in ER+ breast cancer, and there is an FDA-approved treatment targeting it.

Treatment Option: Alpelisib + fulvestrant (SOLAR-1 trial, FDA-approved for PIK3CA-mutated HR+ breast cancer)

DRY_RUN SYNTHETIC DATA — NOT FOR CLINICAL USE. This AI-generated summary must be reviewed and approved by a licensed oncologist before any treatment decisions. Results represent synthetic demonstration data only.

GATA3

Likely Pathogenic

Frameshift variant (somatic, AF=0.38, COSV104733902)

A change in the GATA3 gene was found in your tumor. GATA3 is important for breast cell identity, and changes here are common in ER+ breast cancer. This finding helps confirm the hormone-driven nature of your cancer.

Treatment Option: No direct targeted therapy; supports continued endocrine therapy strategy

MYC

Pathogenic - Copy Number Gain

Amplification (log2 ratio 1.45, ~5 copies, chr8q24)

Extra copies of the MYC gene were found in your tumor cells. MYC drives rapid cell growth and is associated with more aggressive behavior. Investigational BET inhibitor drugs are being studied to target this.

Treatment Option: BET inhibitors (JQ1 class) — investigational; consider clinical trial enrollment

CCND1

Pathogenic - Copy Number Gain

Amplification (log2 ratio 1.62, ~6 copies, chr11q13)

Extra copies of the CCND1 (Cyclin D1) gene were found. This gene controls how fast cells divide. CDK4/6 inhibitors specifically target this pathway and are standard of care in ER+ breast cancer.

Treatment Option: CDK4/6 inhibitors (palbociclib, ribociclib, abemaciclib) — FDA-approved for ER+/HER2- breast cancer

CDKN2A

Pathogenic - Copy Number Loss

Deletion (log2 ratio -1.65, ~1 copy, chr9p21.3)

A copy of the CDKN2A tumor suppressor gene (p16) is missing in your tumor cells. This gene normally acts as a brake on cell division. Its loss may contribute to faster cancer growth and could influence which treatments work best.

Treatment Option: Supports use of CDK4/6 inhibitors; CDKN2A loss may predict response

Your Tumor Environment

Advanced spatial analysis shows how different cells are organized in your tumor:

Tumor Microenvironment

Spatial transcriptomics from 900 Visium spots across 7 tissue regions shows strong ER pathway expression concentrated in tumor core luminal and tumor proliferative regions. ESR1 expression is significantly spatially clustered (Moran's I 0.42, $p < 0.001$), consistent with a luminal A/B ER-driven growth pattern. MKI67 shows weak spatial patterning (Moran's I 0.25), indicating heterogeneous proliferation across the tumor. Immune infiltration (CD8A/CD68) is highest in stroma-immune regions but relatively low in the tumor core — classifying this as an 'immune cold' tumor microenvironment.

Immune Status: • **Low immune activity** - The tumor may be evading your immune system

The spatial analysis shows that your cancer is driven strongly by estrogen signaling, concentrated in the center of the tumor. The immune system cells are mostly around the edges of the tumor rather than inside it — which means the cancer has partially avoided immune attack. This supports continuing hormone-blocking therapy and could be relevant if immunotherapy is considered in the future.

Key Patterns Observed

- ESR1 mean expression 544 counts in tumor_core_luminal vs 16 in stroma — 34-fold enrichment
- PGR mean expression 510 counts in tumor_core_luminal, confirming strong PR+ biology
- GATA3 highest in tumor_core_luminal (664) — supports luminal breast cancer identity

DRY_RUN SYNTHETIC DATA — NOT FOR CLINICAL USE. This AI-generated summary must be reviewed and approved by a licensed oncologist before any treatment decisions. Results represent synthetic demonstration data only.

- CD68 macrophage infiltration elevated in stroma_immune region (278 counts)
- ESR1 Moran's I = 0.42 (significantly clustered, $z=264.7$)
- MKI67 Moran's I = 0.25 (weakly patterned, $z=160.3$) — heterogeneous proliferation

Treatment Options

Based on your test results, your care team may consider these treatments:

Tamoxifen (current) + CDK4/6 Inhibitor (palbociclib, ribociclib, or abemaciclib)

Endocrine
therapy +
targeted

FDA Approved

Expected Response: 60-70% PFS improvement vs endocrine therapy alone

Your current tamoxifen treatment blocks estrogen from fueling your cancer. Adding a CDK4/6 inhibitor (a pill taken daily or in cycles) further slows cancer cell division. This combination is the standard of care for your cancer type and is strongly supported by your CCND1 amplification and CDKN2A deletion.

Why this may help you: ER+/PR+ status plus CCND1 amplification and CDKN2A deletion strongly predict benefit from CDK4/6 inhibitors

Common side effects: Neutropenia (low white blood cells), Fatigue, Nausea, Hot flashes from tamoxifen

DRY_RUN SYNTHETIC DATA — NOT FOR CLINICAL USE. This AI-generated summary must be reviewed and approved by a licensed oncologist before any treatment decisions. Results represent synthetic demonstration data only.

Olaparib (PARP Inhibitor)

Targeted — PARP inhibitor

FDA Approved

Expected Response: 57-60% objective response rate in BRCA-mutated HER2- breast cancer (OlympiAD trial)

Olaparib is a targeted drug (taken as pills twice daily) that blocks a DNA repair pathway your cancer cells rely on because of the BRCA2 mutation. It essentially forces cancer cells to accumulate enough DNA damage to stop growing.

Why this may help you: BRCA2 germline pathogenic frameshift mutation; FDA-approved indication in HER2- metastatic breast cancer with germline BRCA1/2 mutation

Common side effects: Nausea, Fatigue, Anemia, Decreased appetite

Alpelisib + Fulvestrant

Targeted — PI3K inhibitor + endocrine

FDA Approved

Expected Response: 35-40% PFS benefit vs fulvestrant alone (SOLAR-1 trial)

Alpelisib targets the PIK3CA H1047R mutation found in your tumor, blocking a major growth signal. It is combined with fulvestrant (another hormone-blocking drug) to attack the cancer from two angles at once.

Why this may help you: PIK3CA H1047R activating mutation (AF 42%) — the same mutation studied in the SOLAR-1 trial leading to FDA approval

Common side effects: Hyperglycemia (high blood sugar — monitor closely), Rash, Diarrhea, Nausea

DRY_RUN SYNTHETIC DATA — NOT FOR CLINICAL USE. This AI-generated summary must be reviewed and approved by a licensed oncologist before any treatment decisions. Results represent synthetic demonstration data only.

Clinical Trials

You may be eligible for these research studies offering new treatments:

OlympiA: Olaparib as Adjuvant Treatment in Patients with Germline BRCA Mutated High Risk HER2 Negative Primary Breast Cancer

Phase: 3 | **Status:** Active

Trial ID: NCT03796040

This trial uses olaparib as an adjuvant (after surgery) treatment for patients with your exact profile: germline BRCA2 mutation and HER2-negative breast cancer. Ask your oncologist if you qualify.

Contact: clinicaltrials.gov/ct2/show/NCT03796040

INAVO120: Inavolisib + Palbociclib + Fulvestrant in PIK3CA-Mutated HR+/HER2- Locally Advanced or Metastatic Breast Cancer

Phase: 3 | **Status:** Active

Trial ID: NCT05242367

This trial targets your PIK3CA H1047R mutation with a next-generation PI3K inhibitor combined with CDK4/6 inhibitor and fulvestrant — relevant to your tumor's mutational profile.

Contact: clinicaltrials.gov/ct2/show/NCT05242367

DRY_RUN SYNTHETIC DATA — NOT FOR CLINICAL USE. This AI-generated summary must be reviewed and approved by a licensed oncologist before any treatment decisions. Results represent synthetic demonstration data only.

Your Monitoring Plan

Regular check-ups help us track your health and catch any changes early.

Scheduled Tests

Every 6 months	Breast MRI or mammogram Monitor primary tumor response to therapy and detect any new lesions
Every 3 months	Tumor markers: CEA and CA 15-3 Track cancer activity; your current levels (CEA 2.1, CA 15-3 18) are within normal range — any rise could signal progression
Monthly while on CDK4/6 or PARP inhibitor	Complete blood count (CBC) Monitor for bone marrow suppression (low blood counts) from treatment
Monthly if starting alpelisib	Blood glucose (HbA1c) Alpelisib commonly causes high blood sugar that requires management
Annually	Gynecologic exam and pelvic ultrasound BRCA2 germline mutation increases risk of ovarian cancer; annual surveillance recommended
Within 30 days	Genetic counseling referral BRCA2 germline findings have implications for first-degree relatives — siblings and children should be offered testing

⚠ When to Call Your Doctor

Contact your care team right away if you experience:

- New or worsening bone pain (could indicate spread to bones)
- Shortness of breath or chest pain
- Severe or persistent headaches

DRY_RUN SYNTHETIC DATA — NOT FOR CLINICAL USE. This AI-generated summary must be reviewed and approved by a licensed oncologist before any treatment decisions. Results represent synthetic demonstration data only.

- High fever above 38°C/100.4°F while on CDK4/6 inhibitor (sign of infection with low white cells)
- Extreme fatigue or pallor (sign of anemia)

Questions? Your oncology nurse navigator or the 24-hour oncology triage line

For Your Family

The BRCA2 frameshift mutation found in your tumor is a germline (inherited) mutation, meaning you were born with it and it is present in every cell of your body. This mutation can be passed to your children (50% chance per child). Your siblings also have a 50% chance of carrying it. BRCA2 mutations increase the lifetime risk of breast cancer (45-85%), ovarian cancer (10-20%), and some other cancers.

Recommended Steps for Family Members

- Schedule a genetic counseling appointment within the next 4 weeks
- Inform first-degree relatives (parents, siblings, children) about the BRCA2 finding and recommend they speak with their doctor about genetic testing
- Discuss risk-reducing options (prophylactic salpingo-oophorectomy, enhanced surveillance) with your gynecologist
- Consider referral to a hereditary cancer clinic for long-term management

Genetic Counseling Recommended: A genetic counselor can help explain what these results mean for your family members and discuss testing options.

Resources & Support

You don't have to face this alone. These resources can help:



BRCA Foundation

Provides support, education, and genetic testing resources for individuals and families with BRCA1/2 mutations

📞 1-800-292-2776 | 🌐 <https://brcafoundation.org>



Susan G. Komen Breast Cancer Foundation

Helpline, financial assistance programs, and community support for breast cancer patients

📞 1-877-465-6636 | 🌐 <https://www.komen.org>



Facing Our Risk of Cancer Empowered (FORCE)

Dedicated resource for hereditary breast and ovarian cancer — peer navigation, counseling referrals, and BRCA-specific guidance

| 🌐 <https://www.facingourrisk.org>

⚠️ Important Notice

DRY_RUN SYNTHETIC DATA — NOT FOR CLINICAL USE. This AI-generated summary must be reviewed and approved by a licensed oncologist before any treatment decisions. Results represent synthetic demonstration data only.

Data Sources: mockepic (DRY_RUN), fgbio (DRY_RUN), genomic-results, spatialtools, multiomics

Report Version: 1.0

Generated: 2026-03-26 07:07:56

DRY_RUN SYNTHETIC DATA — NOT FOR CLINICAL USE. This AI-generated summary must be reviewed and approved by a licensed oncologist before any treatment decisions. Results represent synthetic demonstration data only.